Software Engineering

UNIT I

Software Engineering

Software engineering is a discipline that applies engineering principles to the design, development, testing, and maintenance of software systems. It aims to produce high-quality software that is reliable, maintainable, and meets user needs.

Software Engineering Concepts

Here are some key concepts in software engineering:

1. Software Development Life Cycle (SDLC)

Definition: A framework that outlines the stages involved in software development from conception to deployment and maintenance.

Phases involved in software development:

a) Requirement Analysis – Gathering and defining what the software needs to accomplish.

Software Engineering Concepts

b) Design – Creating a blueprint for the software system's architecture and components.

c) Implementation (Coding) – Writing the actual code based on the design specifications.

d) Testing – Verifying that the software works as intended and identifying defects.

e) Deployment – Releasing the software to users and setting up the production environment.

f) Maintenance – Updating and fixing the software post-deployment.

Software Engineering Concepts

2. Software Development Models

The phases of SDLC remain the same in all software development processes, but software engineers may choose to implement functions in every phase in different ways through different models.

The most common SDLC models include Waterfall, Agile etc.

Software Engineering Concepts

Waterfall Model: A linear and sequential approach where each phase must be completed before moving to the next. It's simple but can be inflexible to changes.

Agile Methodology: An iterative and incremental approach that emphasizes flexibility, collaboration, and customer feedback. Common frameworks include Scrum and Kanban.

Software Engineering Concepts

3. Requirements Engineering

Definition: The process of defining and managing software requirements. It involves identifying user needs and specifying what the software should do.

Techniques:

Interviews: Directly engaging with stakeholders to gather requirements.

Surveys and Questionnaires: Collecting requirements from a larger group.

Use Cases: Describing how users will interact with the system.

User Stories: Short, simple descriptions of features from an end-user perspective.

Software Engineering Concepts

Stakeholders:

End-users: Those who will directly interact with the software.

Clients/Customers: Individuals or organizations that are paying for or commissioning the software.

Project Managers: Individuals responsible for overseeing the project's progress and ensuring it meets its goals.

Developers: The team responsible for building the software.

Business Analysts: Individuals who analyze business needs and translate them into software requirements.

Testers: Individuals responsible for ensuring the software is working correctly.

Software Engineering Concepts

4. Software Design

Definition: The process of defining the architecture, components, interfaces, and other characteristics of a system.

5. Coding and Implementation

Definition: The actual process of writing and compiling code to build the software.

Software Engineering Concepts

6. Testing

Definition: The process of evaluating software to ensure it meets requirements and is free of defects.

7. Software Maintenance

Definition: Activities required to keep the software operational after deployment, including fixing bugs, updating features, and optimizing performance.

Software Engineering Concepts

8. Software Metrics

Definition: Quantitative measures used to assess various aspects of software development and performance.

Software Engineering Principles

The following set of core principles is fundamental to the practice of software engineering:

Principle 1. Divide and conquer. Stated in a more technical manner, analysis and design should always emphasize separation of concerns (SoCs). A large problem is easier to solve if it is subdivided into a collection of elements (or concerns).

Software Engineering Principles

Principle 2. Understand the use of abstraction. At its core, an abstraction is a simplification of some complex element of a system used to communicate meaning in a single phrase. In software engineering practice, you use many different levels of abstraction, each imparting or implying meaning that must be communicated. In analysis and design work, a software team normally begins with models that represent high levels of abstraction (e.g., a spreadsheet) and slowly refines those models into lower levels of abstraction (e.g., a column or the SUM function).

Software Engineering Principles

Principle 3. Strive (make great effort) for consistency. Whether it's creating an analysis model, developing a software design, generating source code, or creating test cases, the principle of consistency suggests that a familiar context makes software easier to use. For example, consider the design of a user interface for a mobile app. Consistent placement of menu options, the use of a consistent color scheme, and the consistent use of recognizable icons help to create a highly effective user experience.

Software Engineering Principles

Principle 4. Focus on the transfer of information. Software is about information transfer from a database to an end user, from an end user into a graphic user interface (GUI), from an operating system to an application, from one software component to another etc. In every case, information flows across an interface, and this means there are chances for errors, omissions, or ambiguity. The implication of this principle is that you must pay attention to the analysis, design, construction, and testing of interfaces.

Software Engineering Principles

Principle 5. Build software that exhibits effective modularity. That is, each module should focus exclusively on one well-constrained aspect of the system. Additionally, modules should be interconnected in a relatively simple manner to other modules, to data sources, and to other environmental aspects.

Principle 6. Look for patterns. Software engineers use patterns as a means of cataloging and reusing solutions to problems they have encountered in the past. The use of these design patterns can be applied to wider systems engineering and systems integration problems, by allowing components in complex systems to evolve independently.

Software Engineering Principles

Principle 7. When possible, represent thae problem and its solution from several different perspectives. When a problem and its solution are examined from different perspectives, it is more likely that greater insight will be achieved and that errors and omissions will be uncovered.

Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is a structured process that software developers and engineers use to design, develop, test, and deploy software applications. It provides a systematic approach to building software, ensuring quality and efficiency at each stage. The SDLC typically consists of several distinct phases, each with specific objectives and deliverables.

Here's an overview of the main phases in the SDLC:

1. Planning and Requirement Analysis

Objective: Define the project's scope, objectives, and feasibility. Gather and analyze the requirements.

Software Development Life Cycle (SDLC)

Activities:

Stakeholder meetings to gather requirements and expectations.

Feasibility analysis to assess the technical, economic, and operational viability.

Requirement specification to document the functional and non-functional requirements.

Deliverables:

Project plan

Requirement specification document

Feasibility study report

Software Development Life Cycle (SDLC)

2. System Design

Objective: Design the architecture and detailed specifications of the system based on the requirements.

Activities:

High-level design (HLD) to outline the system architecture, modules, and data flow.

Low-level design (LLD) to specify the details of individual components, including data structures, algorithms, and interfaces.

Design of user interfaces, databases, and system security.

Software Development Life Cycle (SDLC)

Deliverables:

High-level design document

Low-level design document

Database design

User interface design

3. Implementation (Coding)

Objective: Convert the design specifications into working code.

Software Development Life Cycle (SDLC)

Activities:

Writing code based on the design documents.

Adhering to coding standards and guidelines.

Using version control systems to manage code changes.

Deliverables:

Source code

Code documentation

Code repositories

Software Development Life Cycle (SDLC)

4. Testing

Objective: Verify that the software meets the specified requirements and identify any defects.

Activities:

Unit testing to verify the functionality of individual components.

Integration testing to check the interactions between integrated modules.

System testing to validate the complete system against requirements.

User Acceptance Testing (UAT) to ensure the system meets user needs and requirements.

Software Development Life Cycle (SDLC)

Deliverables:

Test plans and cases

Test scripts

Test reports

5. Deployment

Objective: Deploy the software to the production environment and make it available to users.

Activities:

Preparing the production environment, including hardware, network, and software configurations.

Software Development Life Cycle (SDLC)

Installing and configuring the software.

Performing final system checks and validation.

Training users and providing documentation.

Deliverables:

Deployed software

Deployment guides

User manuals and training materials

Software Development Life Cycle (SDLC)

6. Maintenance and Support

Objective: Ensure the software continues to function correctly after deployment and meets evolving user needs.

Activities:

Monitoring the system for any issues or bugs.

Providing regular updates, patches, and enhancements.

Handling user support requests and feedback.

Performing preventive maintenance to prevent future issues.

Software Development Life Cycle (SDLC)

Deliverables:

Maintenance logs

Update and patch documentation

User support records

SDLC Models -

There are several SDLC models, each suited to different types of projects and organizational needs. Some of the most common models include:

Waterfall Model: A linear and sequential approach, where each phase must be completed before the next begins. It is simple but less flexible to changes.

Agile Model: An iterative and incremental approach that emphasizes flexibility, collaboration, and customer feedback. It includes frameworks like Scrum and Kanban.

Agile software development model

The Agile software development model is a popular and flexible approach to software system development that emphasizes iterative progress, collaboration, and adaptability.

Here are the key principles and practices that define the Agile methodology:

Key Principles

Individuals and Interactions over Processes and Tools: Agile values the skills, creativity, and communication of team members more than the tools they use.

Working Software over Comprehensive Documentation: The primary measure of progress is working software. Documentation is important, but delivering software that works is more critical.

Agile software development model

Customer Collaboration over Contract Negotiation: Agile emphasizes regular collaboration with customers to understand their needs and adapt the project accordingly.

Responding to Change over Following a Plan: Agile methodologies are designed to be flexible and adaptable, responding to changes even late in development.

Agile software development model

Agile Practices

Iterative Development: Projects are broken down into small iterations or sprints, typically lasting 2-4 weeks. Each iteration involves planning, designing, coding, testing, and reviewing a subset of the final product.

Continuous Feedback: Regular feedback from stakeholders and end-users is incorporated into the development process to ensure the product meets their needs.

Daily Standups: Short, daily meetings where team members discuss what they did the previous day, what they plan to do today, and any blockers they are facing.

User Stories: Requirements are captured as user stories, which describe the desired functionality from the end-user perspective. User stories are prioritized and managed in a product backlog.

Agile software development model

Sprint Planning and Review: Each sprint begins with a planning meeting where the team selects the user stories to work on. At the end of the sprint, a review meeting is held to demonstrate the completed work to stakeholders.

Retrospectives: After each sprint, the team holds a retrospective meeting to discuss what went well, what didn’t, and how processes can be improved in the next sprint.

Cross-Functional Teams: Agile teams are typically composed of members with different skills (e.g., developers, testers, designers) to foster collaboration and ensure all aspects of the product are addressed.

Agile software development model

Popular Agile Frameworks –

Scrum: A framework within which people can address complex adaptive problems, while productively and creatively delivering products of the highest possible value. Scrum includes roles (Scrum Master, Product Owner, Development Team), events (Sprint, Daily Scrum, Sprint Review, Sprint Retrospective), and artifacts (Product Backlog, Sprint Backlog, Increment).

Kanban: Focuses on visualizing the flow of work, limiting work in progress, and maximizing efficiency. It uses a Kanban board to visualize tasks and their status.

Scrum framework for software development

The Scrum framework is one of the most widely adopted Agile methodologies for managing and completing complex projects. It provides a structured approach to iterative development through defined roles, events, and artifacts.

Key Roles

Product Owner:

1. Represents the stakeholders and the voice of the customer.

2. Responsible for defining and prioritizing the product backlog.

3. Ensures the team is working on the most valuable features.

Scrum framework for software development

Scrum Master:

1. Facilitates the Scrum process and ensures that the team adheres to Scrum practices.

2. Helps resolve impediments that the team encounters.

3. Acts as a coach for the team, ensuring that Agile principles are followed.

Development Team:

1. A cross-functional group of professionals (developers, testers, designers) responsible for delivering potentially shippable increments of the product.

2. Self-organizing and responsible for managing their work.

Scrum framework for software development

Key Events

Sprint:

A time-boxed iteration, typically 2-4 weeks, during which a potentially shippable product increment is created.

Consists of planning, execution, review, and retrospective phases.

Sprint Planning:

A meeting held at the beginning of each sprint to determine what can be delivered in the upcoming sprint and how that work will be achieved.

The Product Owner presents the top items from the product backlog, and the team selects the items they can commit to.

Scrum framework for software development

Daily Scrum (Daily Standup):

A short, daily meeting (usually 15 minutes) where the development team synchronizes their work and plans for the next 24 hours.

Team members answer three questions: What did I do yesterday? What will I do today? Are there any impediments in my way?

Sprint Review:

Held at the end of the sprint to inspect the increment and adapt the product backlog if needed.

The development team demonstrates what has been completed during the sprint.

Stakeholders provide feedback.

Scrum framework for software development

Sprint Retrospective:

A meeting held after the sprint review and before the next sprint planning to reflect on the sprint.

The team discusses what went well, what didn’t, and how processes can be improved.

Key Artifacts

Product Backlog:

An ordered list of everything that is known to be needed in the product.

Managed by the Product Owner and continuously refined.

Scrum framework for software development

Sprint Backlog:

A list of items selected from the product backlog to be completed in the current sprint, along with a plan for delivering the product increment and achieving the sprint goal.

Increment:

The sum of all the product backlog items completed during a sprint and all previous sprints.

Must be in a usable condition regardless of whether the Product Owner decides to release it.

Backlog Refinement: The Product Owner and development team regularly refine the product backlog to ensure that it is well-defined and prioritized.

Scrum framework for software development

Sprint Planning: At the start of each sprint, the team selects items from the product backlog to move to the sprint backlog and defines a sprint goal.

Daily Scrum: The team meets daily to discuss progress, plan the day's work, and identify any impediments.

Sprint Execution: The development team works on the tasks in the sprint backlog to create a potentially shippable product increment.

Sprint Review: At the end of the sprint, the team presents the increment to stakeholders for feedback.

Sprint Retrospective: The team reflects on the sprint and identifies improvements for the next sprint.

Kanban framework for software development

The Kanban framework is a visual process management system that helps teams visualize their work, limit work in progress, and maximize efficiency or flow. Unlike Scrum, which has predefined roles and events, Kanban is more flexible and can be applied to various existing workflows without necessitating drastic changes.

Key Principles

Visualize Work: Use a Kanban board to represent work items and their status. This visualization helps the team understand the flow of work and identify bottlenecks.

Kanban framework for software development

Limit Work in Progress (WIP): Restrict the number of work items in progress at any stage of the workflow to ensure focus and efficiency.

Manage Flow: Continuously monitor and manage the flow of work items from start to finish to optimize throughput.

Make Process Policies Explicit: Clearly define and communicate the process policies so that everyone understands how work should be done.

Implement Feedback Loops: Regularly review and adapt processes based on feedback to improve continuously.

Kanban framework for software development

Improve Collaboratively, Evolve Experimentally: Use data and collaborative discussion to drive continuous improvement in small, incremental changes.

Key Components

Kanban Board: The central tool in Kanban, which visually represents the workflow. It typically includes columns such as "To Do," "In Progress," and "Done," but can be customized to fit the team's process.

Cards: Represent individual work items or tasks. Each card includes information about the task, such as a description, assignee, due date, and other relevant details.

Kanban framework for software development

WIP Limits: Set maximum limits for the number of work items in each column (e.g., no more than 3 items in "In Progress") to prevent overloading and to focus on finishing tasks.

Visualize Workflow: Map out the entire workflow on the Kanban board, from initial request to delivery. Break down the workflow into distinct stages or columns.

Create and Move Cards: Each work item is represented by a card that moves across the board from left to right as it progresses through different stages.

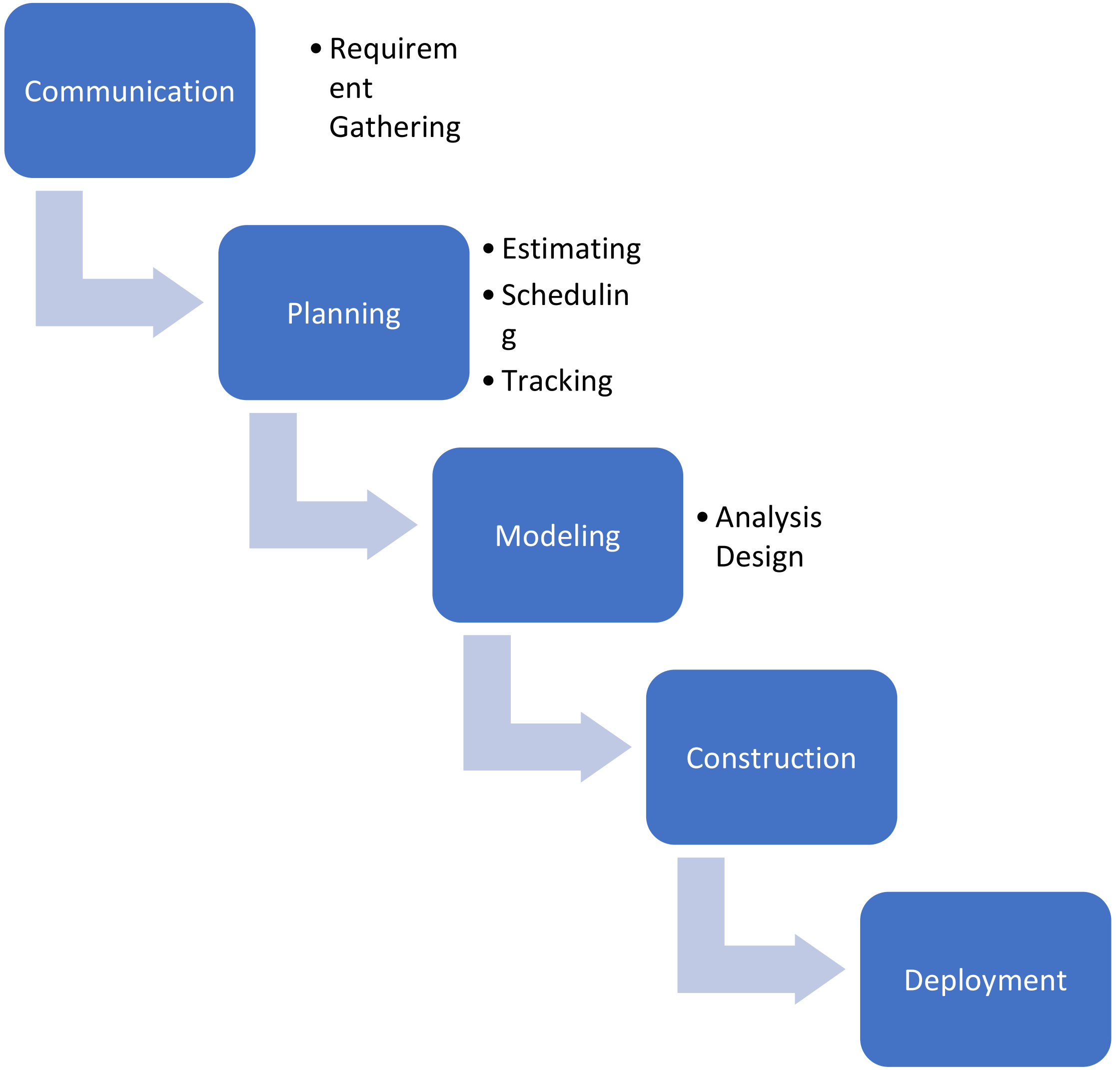
Set WIP Limits: Establish WIP limits for each column to ensure that the team does not take on too much work at once, maintaining focus and productivity.

Kanban framework for software development

Monitor and Manage Flow: Regularly review the board to identify bottlenecks, track progress, and manage the flow of work. Ensure that work items are moving smoothly from one stage to the next.

Continuous Improvement: Use metrics such as lead time (time from task initiation to completion) and cycle time (time from start of work to completion) to identify areas for improvement. Hold regular meetings to discuss process improvements.

waterfall model



Waterfall Model

The Waterfall Model is a linear and sequential approach to software development. It is one of the earliest software development methodologies and is named "Waterfall" because the process flows downwards through distinct phases, similar to a waterfall.

Phases of the Waterfall Model

1. Requirement Analysis

- Gather and document all software requirements.
- Finalized requirements are not expected to change.

Waterfall Model

2. System Design

- Define how the system will meet the requirements.
- Includes architectural and detailed design specifications.

3. Implementation (Coding)

- Developers write code based on the design documents.
- Each component is built and tested individually.

4. Integration and Testing

- All components are integrated.
- The complete system is tested for defects and bugs.

Waterfall Model

5. Deployment

- The final product is delivered to the customer or end-users.

6. Maintenance

- Bug fixes, updates, and enhancements are made after deployment.

Waterfall Model

Advantages -

- Simple and easy to understand and manage.
- Clear milestones and well-defined stages.
- Suitable for small projects with well-understood requirements.

Disadvantages -

- Inflexible to changes (difficult to go back to a previous stage).
- Poor model for long-term or complex projects.
- Late discovery of issues, since testing happens after development.
- Doesn’t accommodate evolving requirements well.

Requirements Engineering

Requirements Engineering (RE) is a critical process in software development focused on identifying, documenting, and maintaining the needs and functionalities of a software system. It ensures that the software delivers what stakeholders expect.

1. Requirements Elicitation

The process of gathering requirements from stakeholders (e.g., customers, users, business managers). It's about discovering the real needs.

Requirements Engineering

Objectives:

Understand the problem domain.

Identify the stakeholders.

Gather relevant requirements (functional and non-functional).

Techniques:

Technique	Description
Interviews	One-on-one discussions with stakeholders to gather insights.

Requirements Engineering

Questionnaires/Surveys	Useful for gathering data from many users.
Workshops	Collaborative sessions with stakeholders and developers.
Observation	Watching users in their work environment to uncover implicit needs.
Document Analysis	Studying existing systems, business documents, or manuals.
Brainstorming	Generating ideas quickly with a group of stakeholders.

Requirements Engineering

Use Case/Scenario Analysis	Writing detailed narratives to describe how users will interact with the system.
Prototyping	Building mockups to clarify vague requirements.

2. Requirements Analysis

Examining, organizing, and prioritizing the gathered requirements to understand conflicts, dependencies, and feasibility.

Requirements Engineering

Objectives:

Detect and resolve inconsistencies.

Categorize requirements.

Assess feasibility and risks.

Prioritize requirements.

Techniques:

Technique	Description
Modeling (UML, DFDs, ER Diagrams)	Visually represent requirements and system behavior.

Requirements Engineering

Requirements Classification	Grouping into functional, non-functional, domain, user, etc.
Conflict Resolution	Negotiating between stakeholders to resolve contradictions.
Prioritization	Rank requirements by importance.
SWOT Analysis	Evaluate strengths, weaknesses, opportunities, and threats.
Gap Analysis	Identify the difference between current and desired systems.

Requirements Engineering

3. Requirements Specification

Documenting the analyzed and agreed-upon requirements clearly and precisely in a format understandable by both technical and non-technical stakeholders.

Objectives:

Create a Software Requirements Specification (SRS).

Ensure clarity, completeness, consistency, and

Requirements Engineering

Techniques:

Technique	Description
Natural Language Specification	Writing requirements in plain, structured English.
Use Cases & User Stories	Describing system functionality from a user's perspective.
SRS Document (IEEE 830 Standard)	Standard format for formal documentation of all requirements.

Requirements Engineering

4. Requirements Validation

Ensuring that the documented requirements accurately reflect stakeholder needs and are feasible, correct, and complete before development begins.

Objectives:

Catch errors early.

Get stakeholder approval.

Ensure requirements are implementable.

Requirements Engineering

Techniques:

Technique	**Description**
Reviews/Walkthroughs	Manual checking of the requirements document with stakeholders.
Prototyping	Build basic working models to confirm requirements.
Test Case Generation	Creating test cases to validate requirements are testable.
Model Validation	Simulate or animate models (like use case diagrams).

Requirements Engineering

Checklists Systematically verify requirement quality (completeness, clarity, etc.).

Software requirements specification (SRS)

A software requirements specification (SRS) is a description of a software system to be developed. It lays out functional and non-functional requirements and may include a set of use cases that describe user interactions that the software must provide.

Software requirements specification (SRS)

Introduction

Purpose of this Document – At first, main aim of why this document is necessary and what's purpose of document is explained and described.

Scope of this document – In this, overall working and main objective of document and what value it will provide to customer is described and explained. It also includes a description of development cost and time required.

Overview – In this, description of product is explained. It's simply summary or overall review of product.

Software requirements specification (SRS)

General description

In this, general functions of product which includes objective of user, a user characteristic, features, benefits, its importance. It also describes features of user community.

Functional Requirements

In this, possible outcome of software system which includes effects due to operation of program is fully explained. All functional requirements which may include calculations, data processing, etc. are placed in a ranked order. Functional requirements specify the expected behavior of the system-which outputs should be produced from the given inputs.

Software requirements specification (SRS)

They describe the relationship between the input and output of the system. For each functional requirement, detailed description all the data inputs and their source, the units of measure, and the range of valid inputs must be specified.

Interface Requirements

In this, software interfaces which mean how software program communicates with each other or users either in form of any language, code, or message are fully described and explained. Examples can be shared memory, data streams, etc.

Software requirements specification (SRS)

Performance Requirements

In this, how a software system performs desired functions under specific condition is explained. It also explains required time, required memory, maximum error rate, etc. All the requirements relating to the performance characteristics of the system must be clearly specified. There are two types of performance requirements: static and dynamic.

Software requirements specification (SRS)

Design Constraints

In this, constraints limitation or restriction are specified and explained for design team. Examples may include use of a particular algorithm, hardware and software limitations, etc. There are a number of factors in the client's environment that may restrict the choices of a designer leading to design constraints such factors include standards that must be followed resource limits, operating environment, reliability and security requirements and policies that may have an impact on the design of the system.

Software requirements specification (SRS)

Non-Functional Attributes

In this, non-functional attributes are explained that are required by software system for better performance. An example may include Security, Portability, Reliability, Reusability, Application compatibility, Data integrity, Scalability capacity, etc.

Preliminary Schedule and Budget

In this, initial version and budget of project plan are explained which include overall time duration required and overall cost required for development of project.

Software requirements specification (SRS)

Appendices

In this, additional information like references from where information is gathered, definitions of some specific terms, acronyms, abbreviations, etc. are given and explained.

Uses of SRS document

- Development team require it for developing product according to the need.
- Test plans are generated by testing group based on the describe external behaviour.
- Maintenance and support staff need it to understand what the software product is supposed to do.

Software requirements specification (SRS)

- Project manager base their plans and estimates of schedule, effort and resources on it.
- customer rely on it to know that product they can expect.
- As a contract between developer and customer.

Software requirements specification (SRS)

SRS's characteristics include:

Correct
Unambiguous
Complete
Consistent
Ranked for importance and/or stability
Verifiable
Modifiable
Traceable